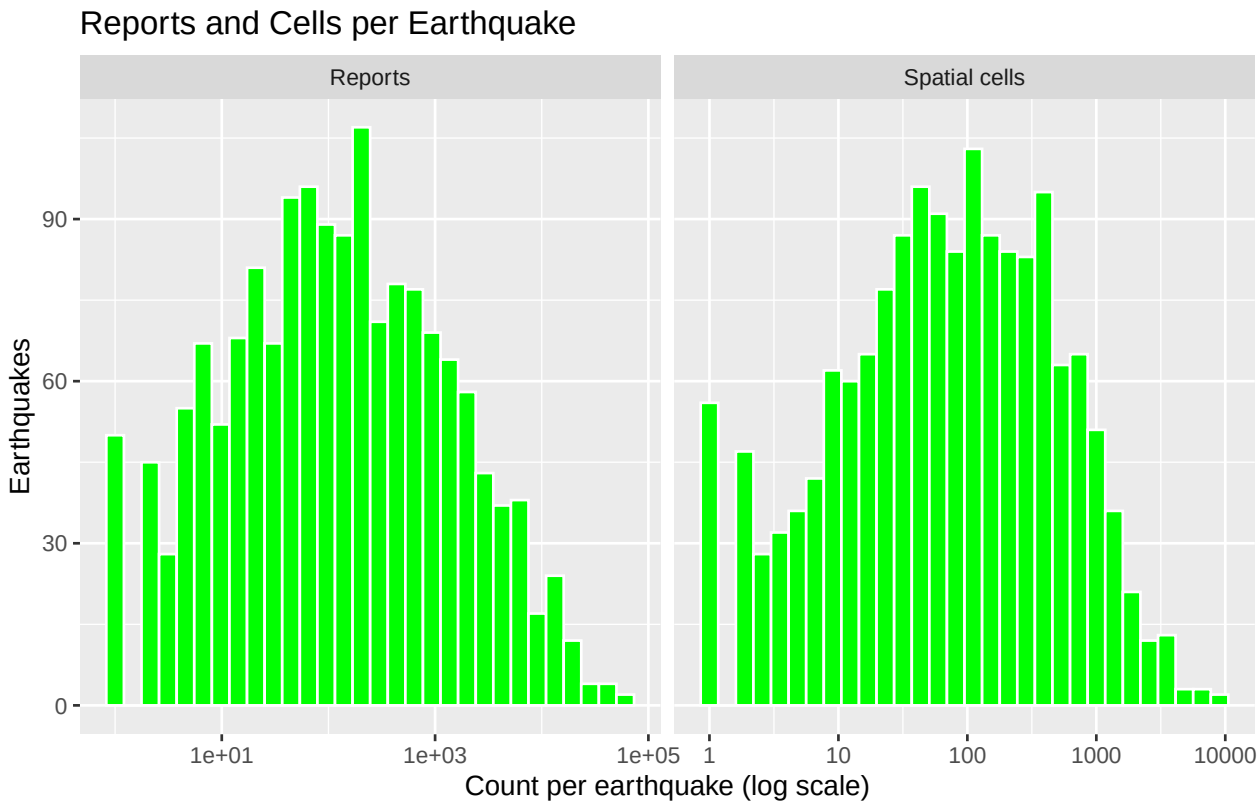
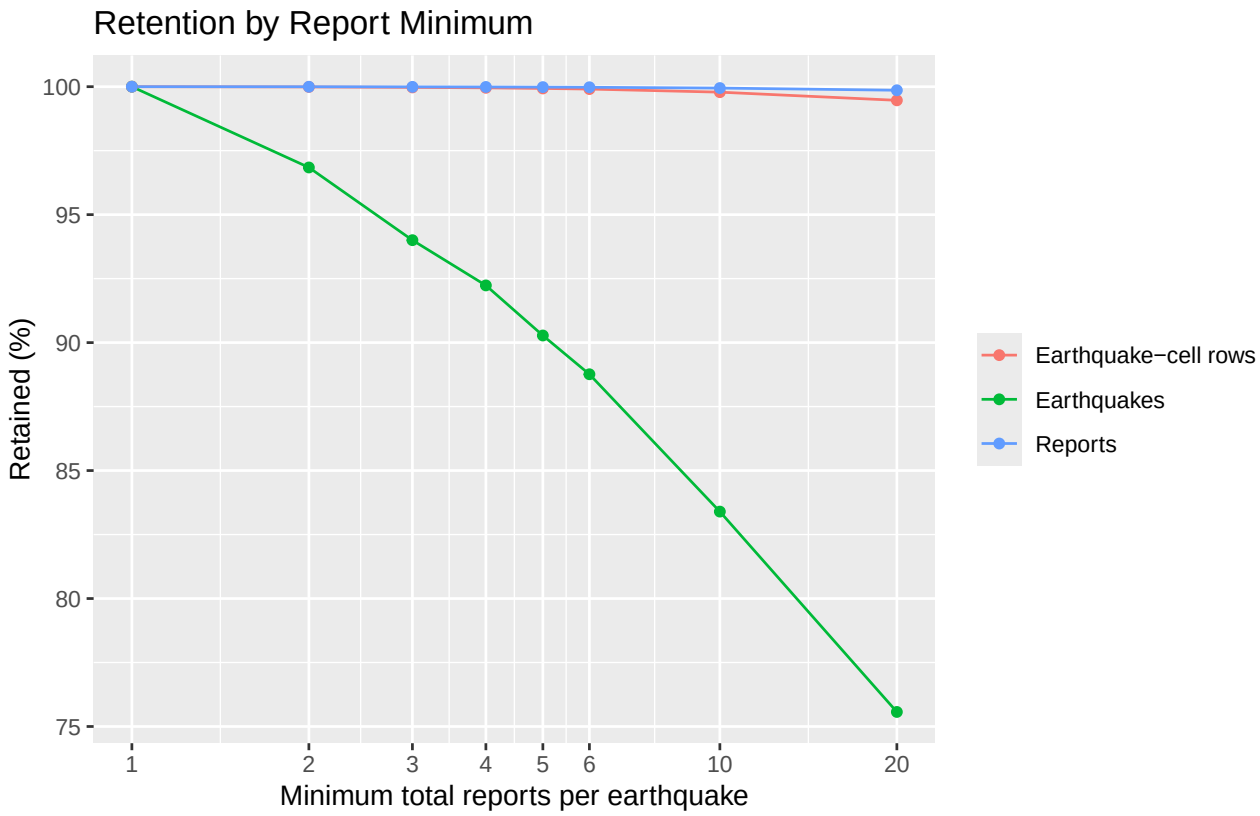
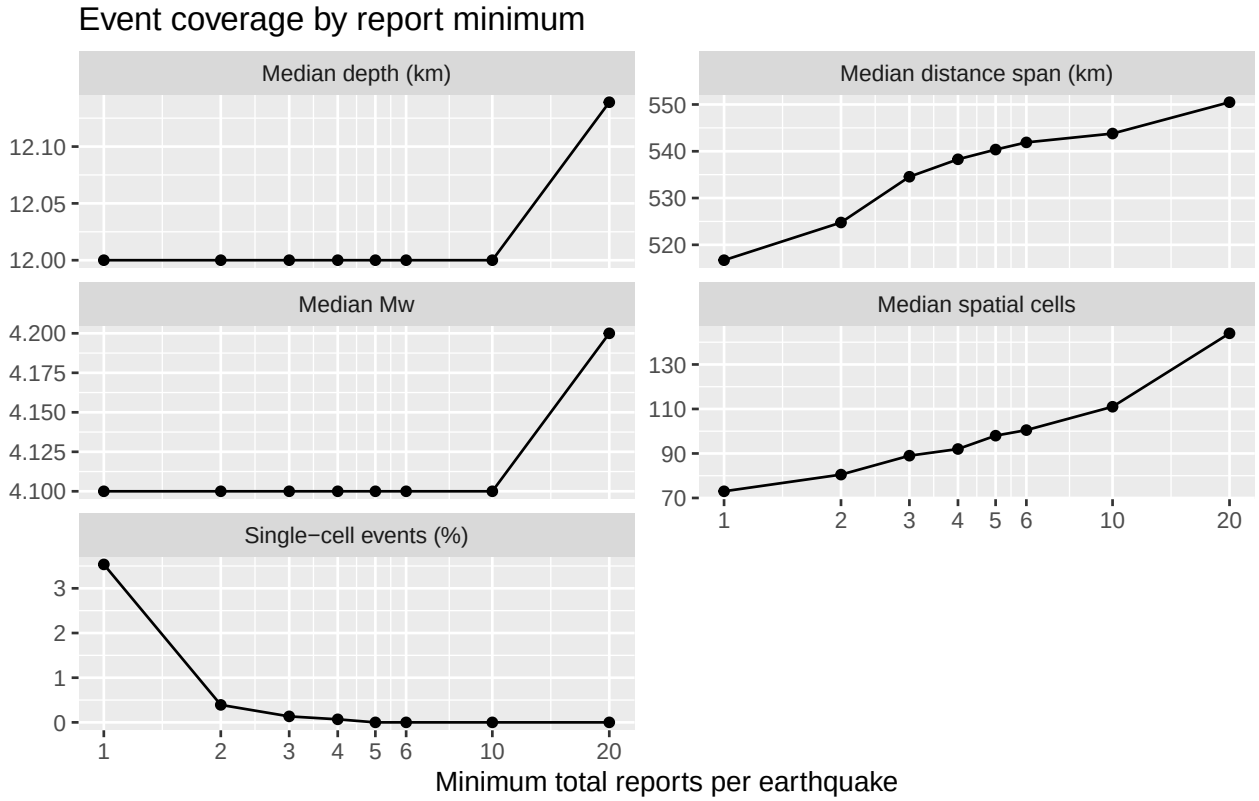
Exploratory Data Analysis

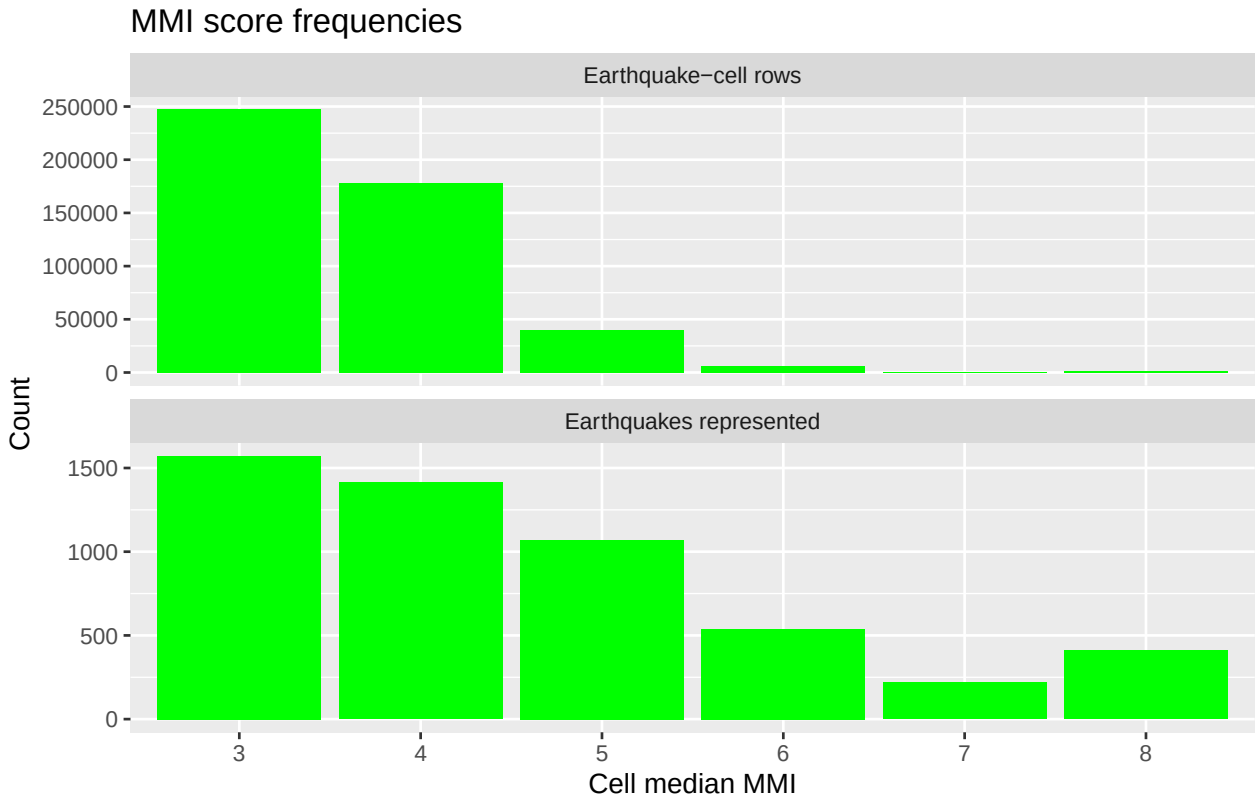
Thomas Rautao

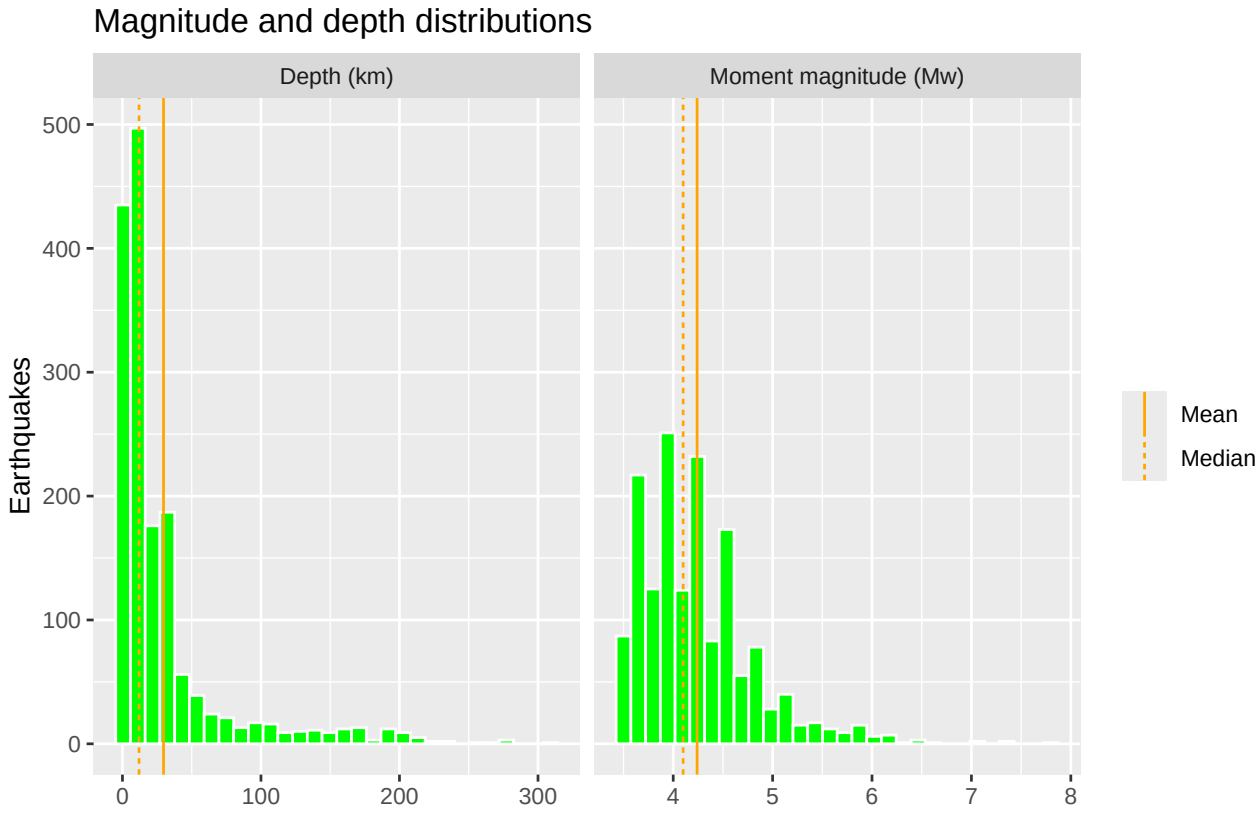
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Warning: package 'arrow' was built under R version 4.6.1
##
## Attaching package: 'arrow'
##
## The following object is masked from 'package:lubridate':
##
##     duration
##
## The following object is masked from 'package:utils':
##
##     timestamp
## Warning: package 'sf' was built under R version 4.6.1
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
##
## Attaching package: 'rnaturalearthdata'
##
## The following object is masked from 'package:rnaturalearth':
##
##     countries110
##
## terra 1.9.34
##
## Attaching package: 'terra'
##
## The following object is masked from 'package:arrow':
##
##     buffer
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

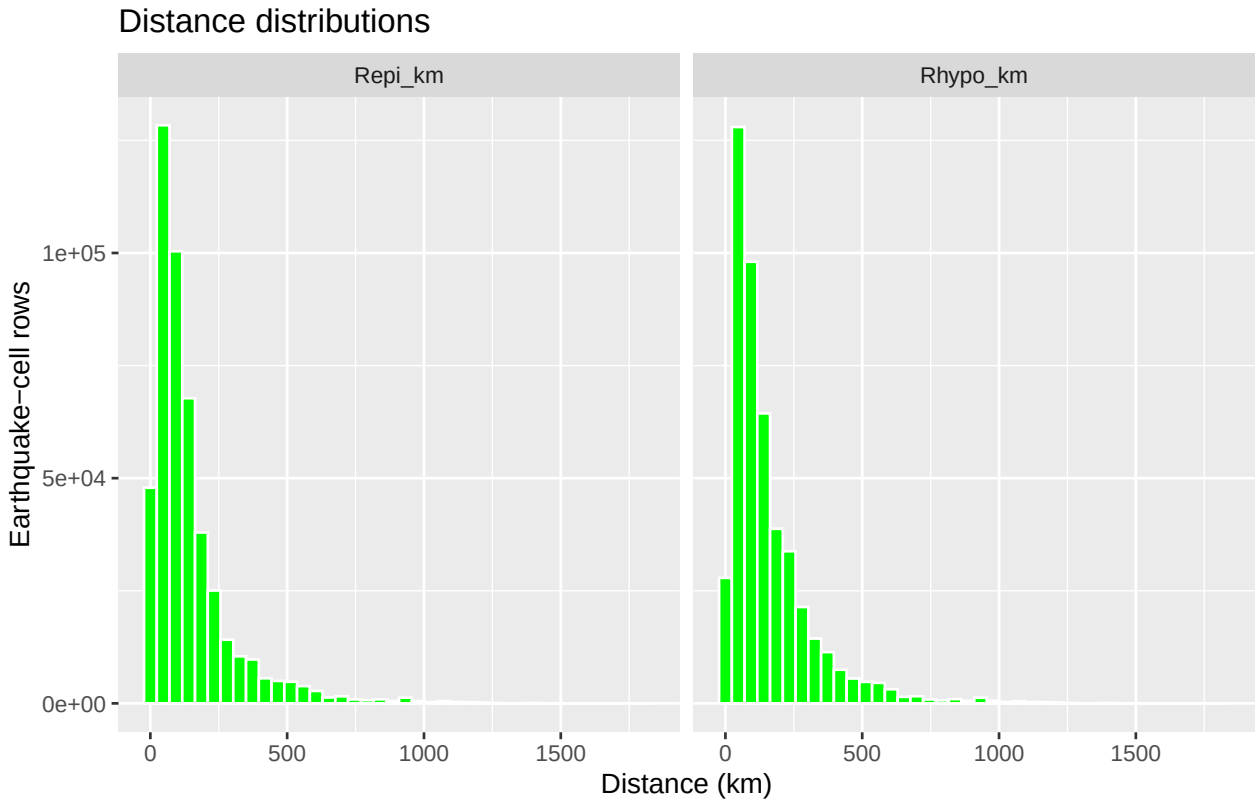

Reports and Cells per Earthquake
Reports
Spatial cells
Earthquakes
90
60
30
0
1e+01
1e+03
1e+05
1
10
100
1000
10000
Count per earthquake (log scale)

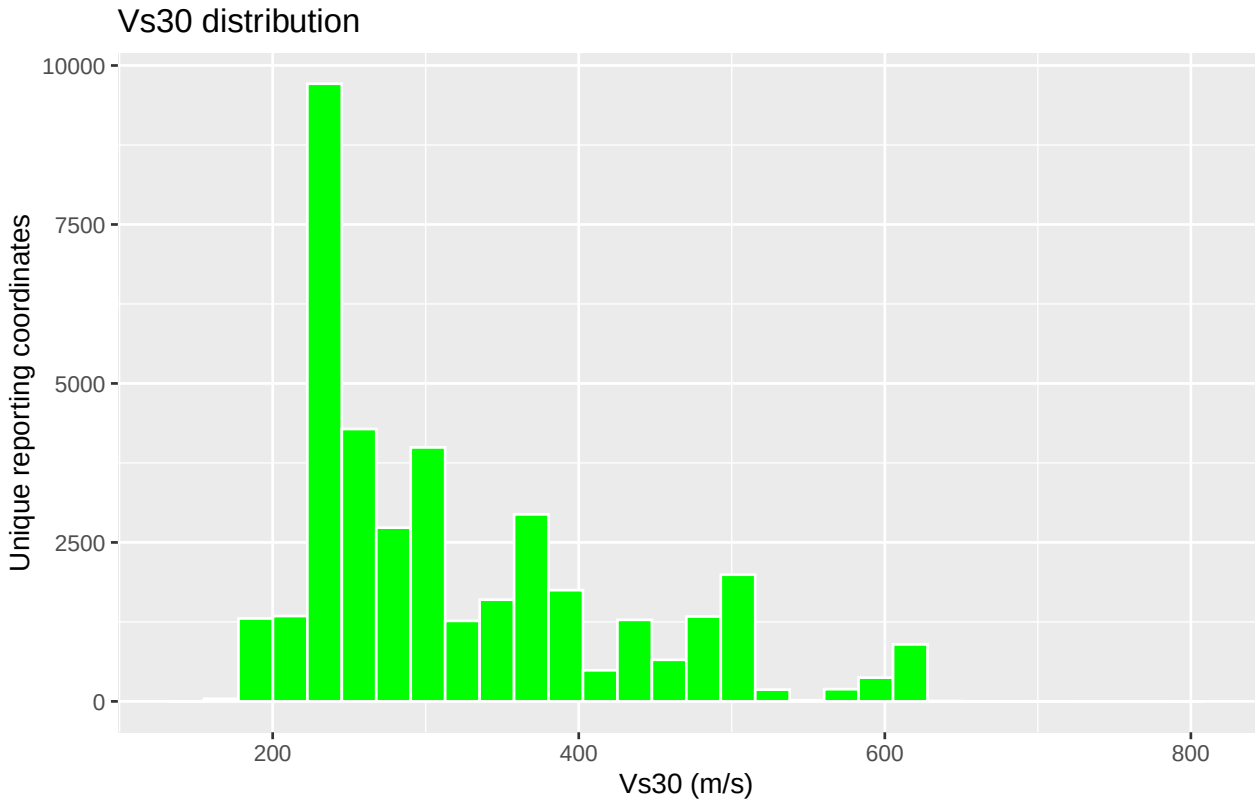

Retention by Report Minimum
Retained (%)
100
95
90
85
80
75
1
2
3
4
5
6
10
20
Minimum total reports per earthquake
Earthquake−cell rows
Earthquakes
Reports

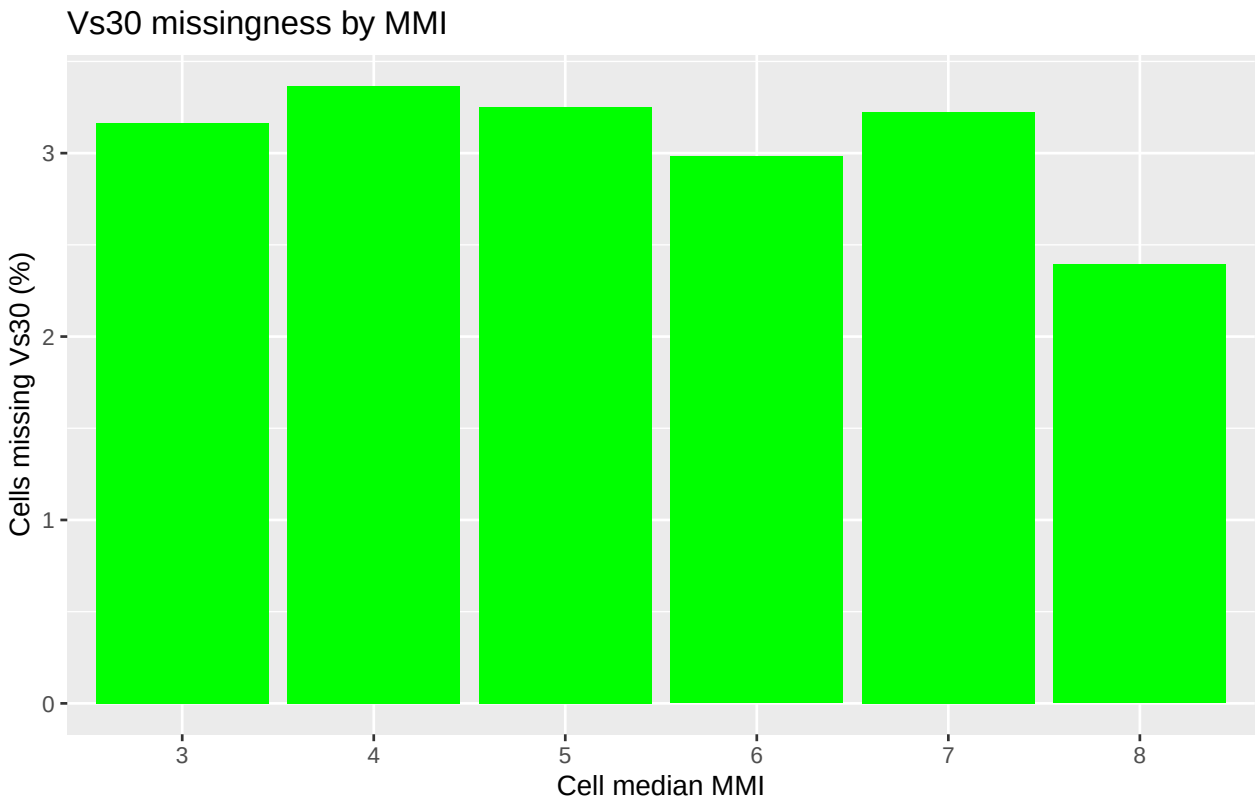

Event coverage by report minimum
Median depth (km)
12.10
12.05
12.00
Median distance span (km)
550
540
530
520
Median Mw
4.200
4.175
4.150
4.125
4.100
Median spatial cells
130
110
90
70
Single−cell events (%)
3
2
1
0
1
2
3
4
5
6
10
20
Minimum total reports per earthquake

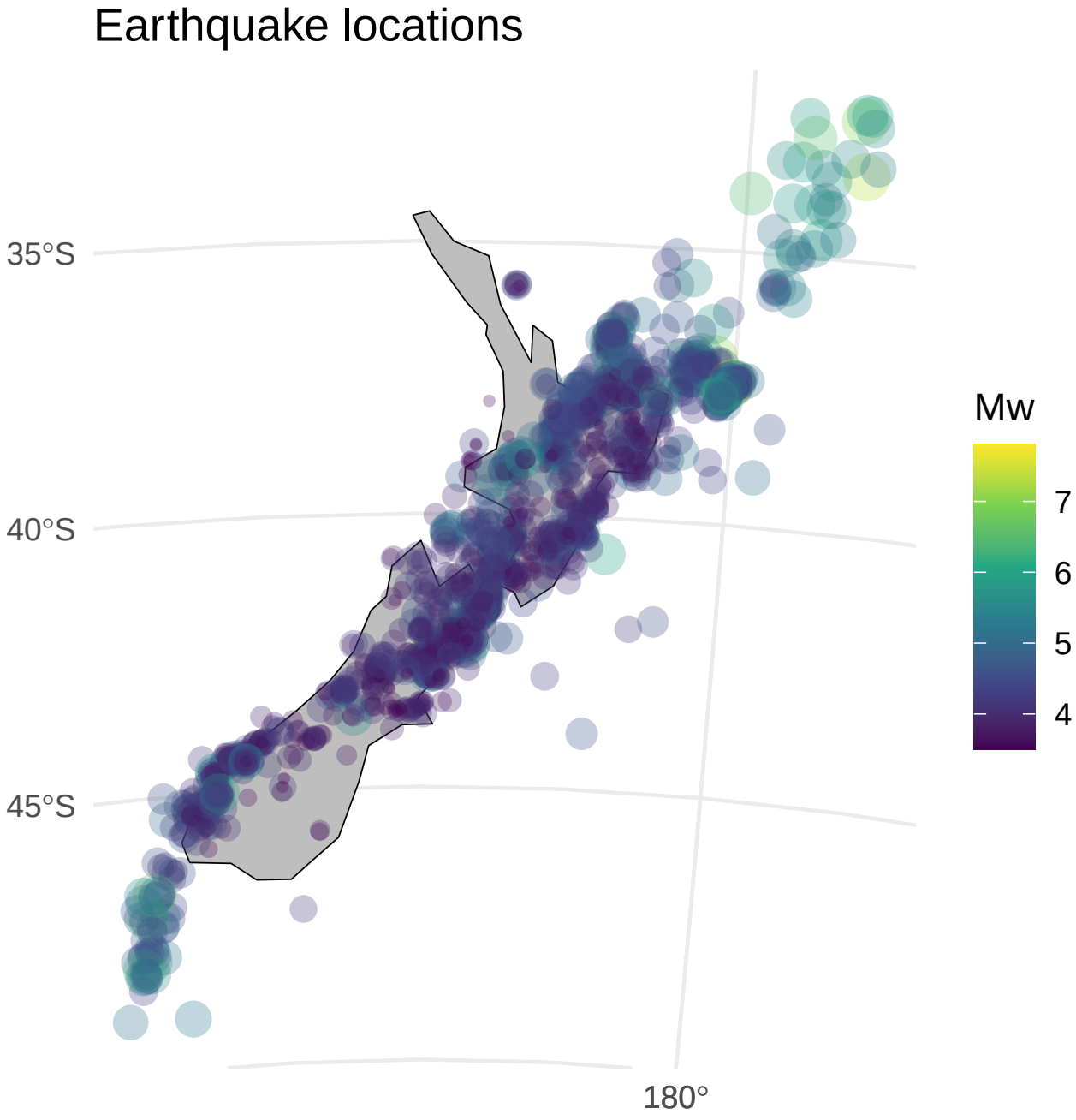

MMI score frequencies
Earthquake−cell rows
250000
200000
150000
100000
50000
0
Earthquakes represented
1500
1000
500
0
Count
3
4
5
6
7
8
Cell median MMI

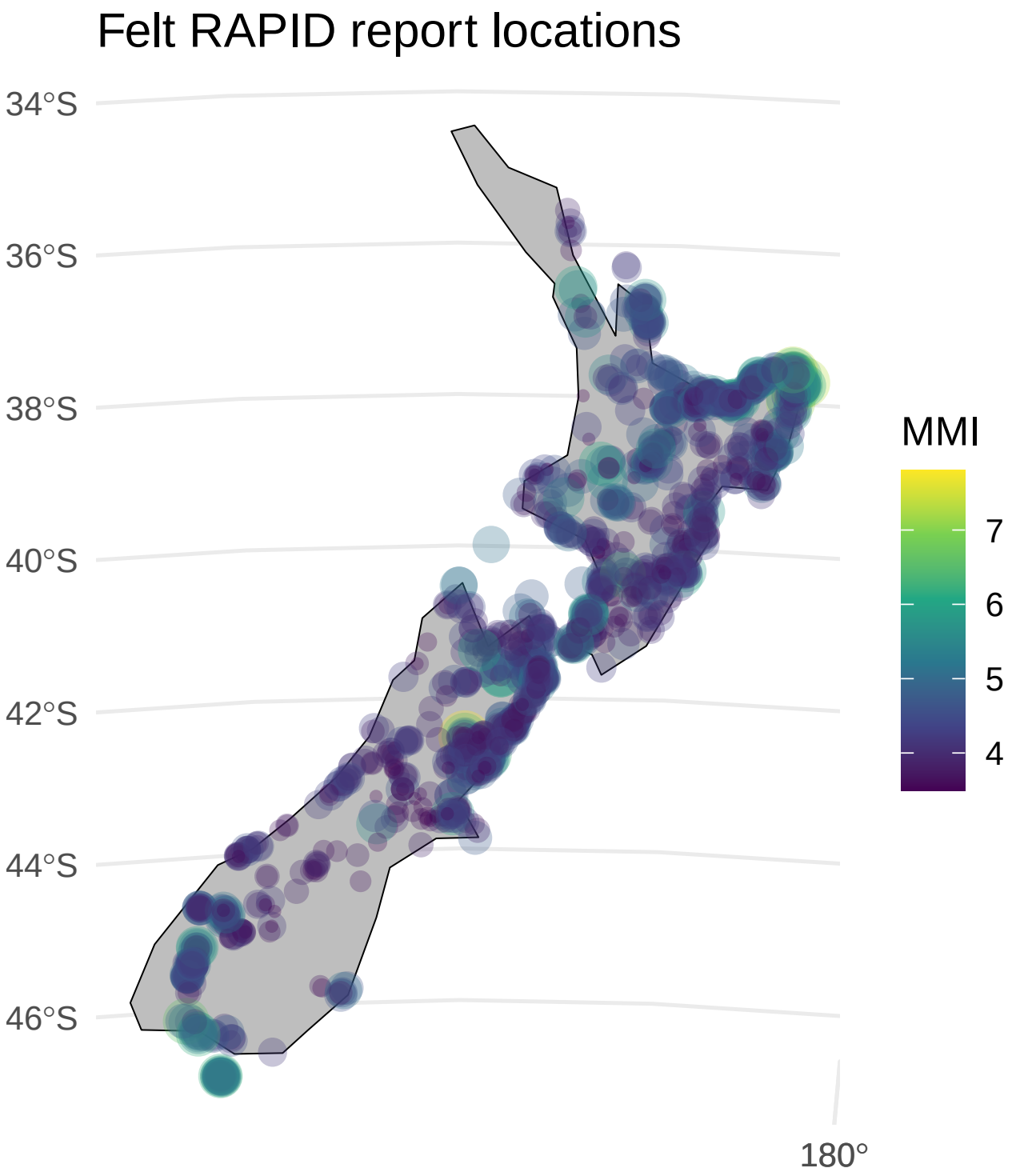

Magnitude and depth distributions
Depth (km)
Moment magnitude (Mw)
Earthquakes
500
400
300
200
100
0
0
100
200
300
4
5
6
7
8
Mean
Median

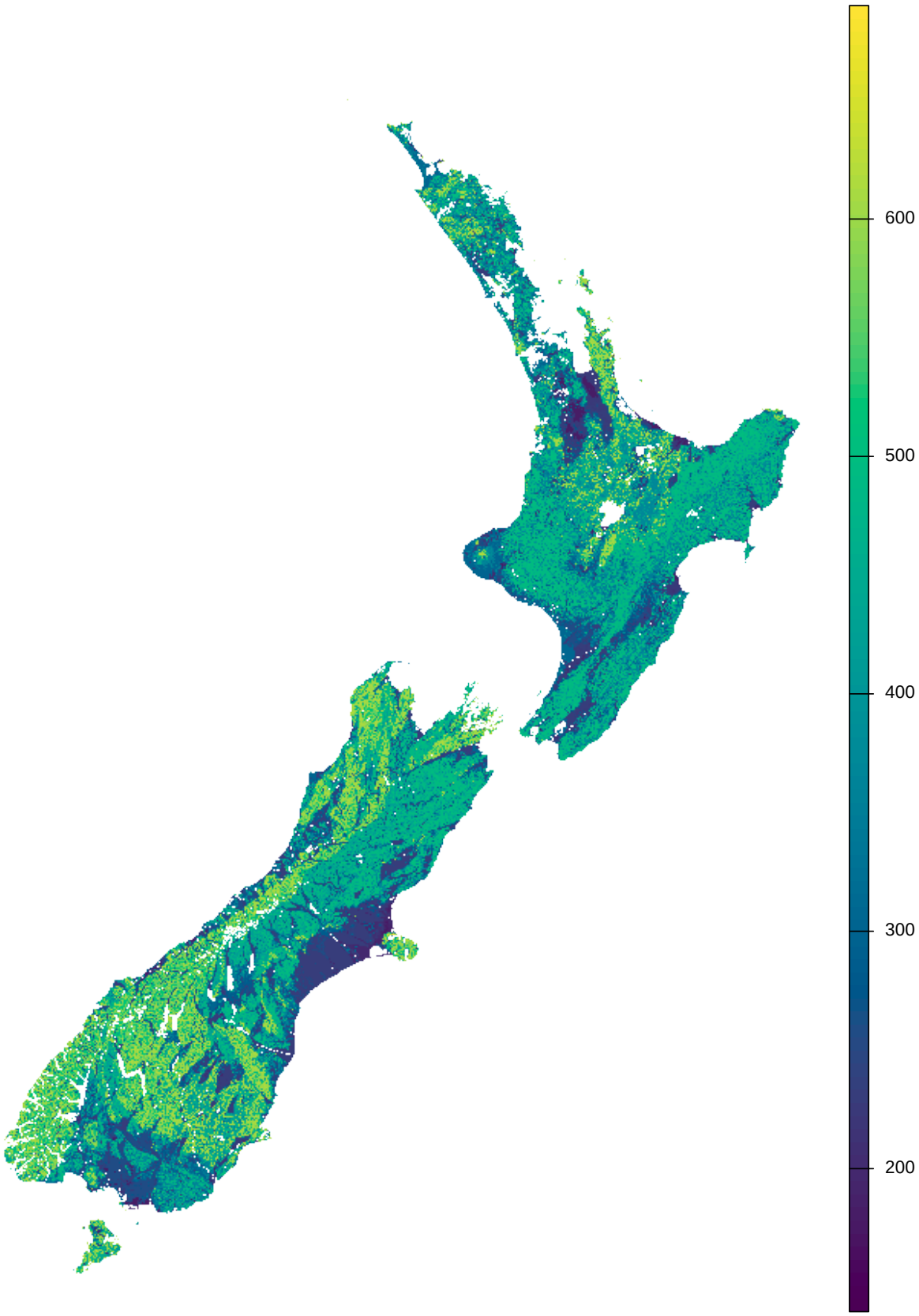

Distance distributions
Repi_km
Rhypo_km
Earthquake−cell rows
1e+05
5e+04
0e+00
0
500
1000
1500
Distance (km)


Vs30 distribution
Unique reporting coordinates
10000
7500
5000
2500
0
200
400
600
800
Vs30 (m/s)

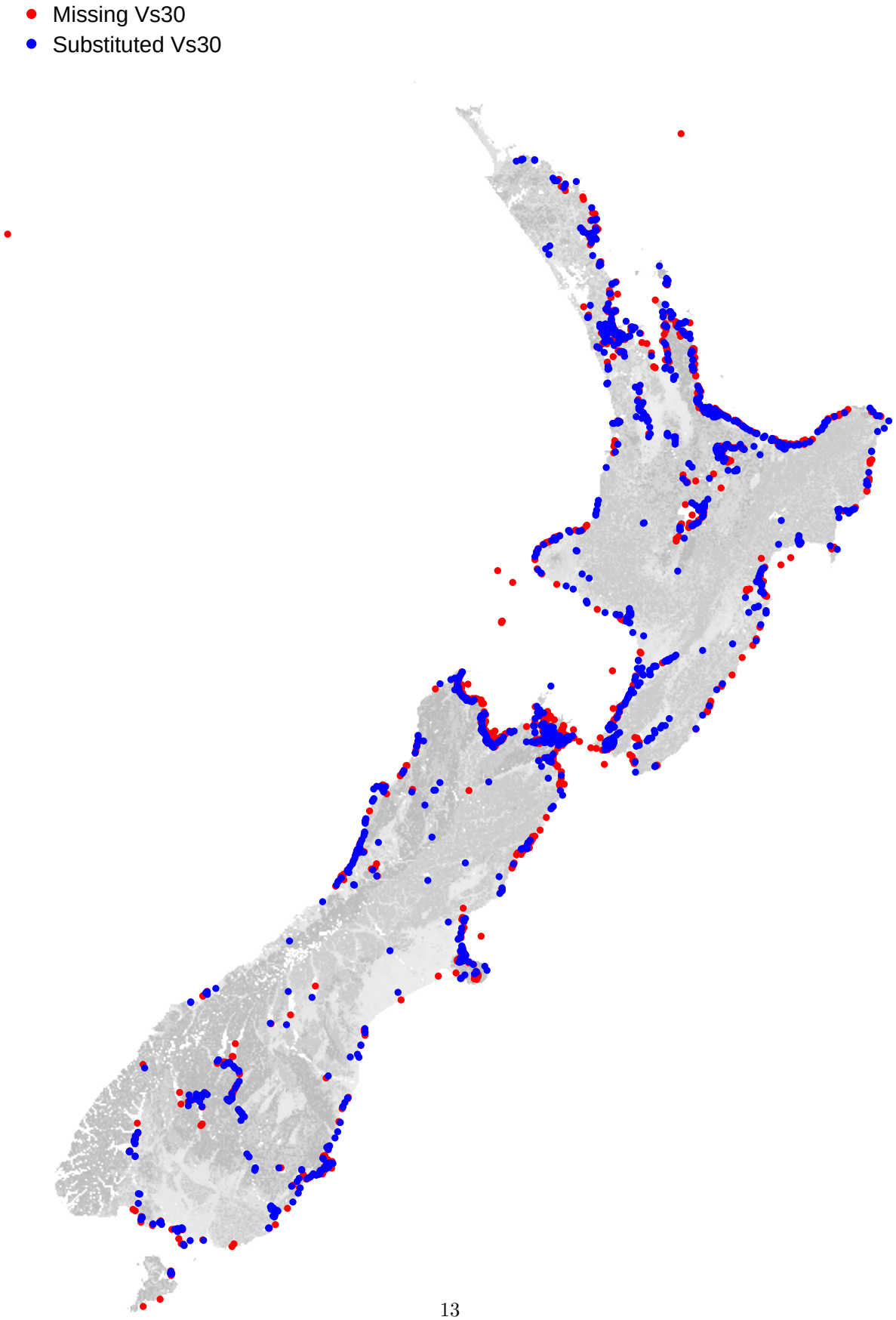

Vs30 missingness by MMI
Cells missing Vs30 (%)
0
1
2
3
3
4
5
6
7
8
Cell median MMI


Earthquake locations
35°S
40°S
45°S
180°
Mw
7
6
5
4


Felt RAPID report locations
34°S
36°S
38°S
40°S
42°S
44°S
46°S
180°
MMI
7
6
5
4


New Zealand Vs30 (m/s)
600
500
400
300
200

Missing and Substituted Vs30 Values



```
##
##     Missing    Original Substituted
##        1403       37246        1147
```